

Comparing Smooth Basis Types

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Overview

GAMs represent smooth functions as linear combinations of **basis functions**. The choice of basis affects the shape of the fitted smooth, computational cost, and numerical properties. `mgcv` supports several basis types:

Code	Basis	Description
"tp"	Thin plate regression spline	Default. Optimal in a certain sense; no knot placement needed
"cr"	Cubic regression spline	Cubic spline with knots at data quantiles; efficient for 1D
"ps"	P-spline	B-spline basis with difference penalty
"gp"	Gaussian process	Squared-exponential covariance as a basis

This vignette fits the same data with each basis and compares the results.

Setup

```
library(mgcv)
```

Loading required package: nlme

This is mgcv 1.9-3. For overview type 'help("mgcv-package")'.

```
library(gratia)
library(ggplot2)
```

Loading data

We load $n = 300$ observations generated from a function with both broad and fine-scale structure:

$$y_i = \sin(2\pi x_i) + 0.5 \sin(6\pi x_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, 0.5^2)$$

```
dat <- read.csv("../data.csv")
x <- dat$x
y <- dat$y
n <- nrow(dat)
f_true <- sin(2 * pi * x) + 0.5 * sin(6 * pi * x)
df <- data.frame(x = x, y = y)
```

Fitting models with different bases

We fit the same formula with each basis type, using $k = 20$ basis functions:

```
bases <- c("tp", "cr", "ps", "gp")
models <- list()

for (bs in bases) {
  models[[bs]] <- gam(y ~ s(x, k = 20, bs = bs), data = df, method = "REML")
}
```

Comparing EDF and deviance

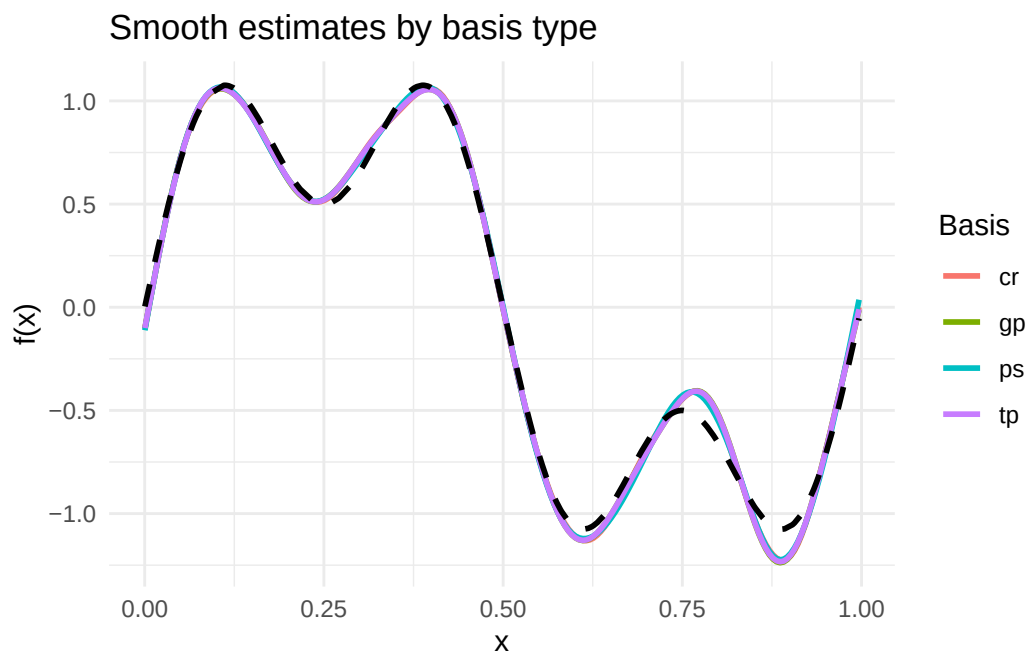
```
results <- data.frame(  
  Basis = bases,  
  EDF = sapply(models, function(m) round(sum(pen.edf(m)), 2)),  
  Deviance = sapply(models, function(m) round(deviance(m), 2)),  
  Dev_Explained = sapply(models, function(m) round(summary(m)$dev.expl * 100, 1))  
)  
print(results)
```

	Basis	EDF	Deviance	Dev_Explained
tp	tp	12.92	64.44	75.1
cr	cr	12.78	64.49	75.1
ps	ps	11.66	64.63	75.1
gp	gp	12.62	64.42	75.2

Comparing smooth estimates

We evaluate each smooth on the same grid and compare:

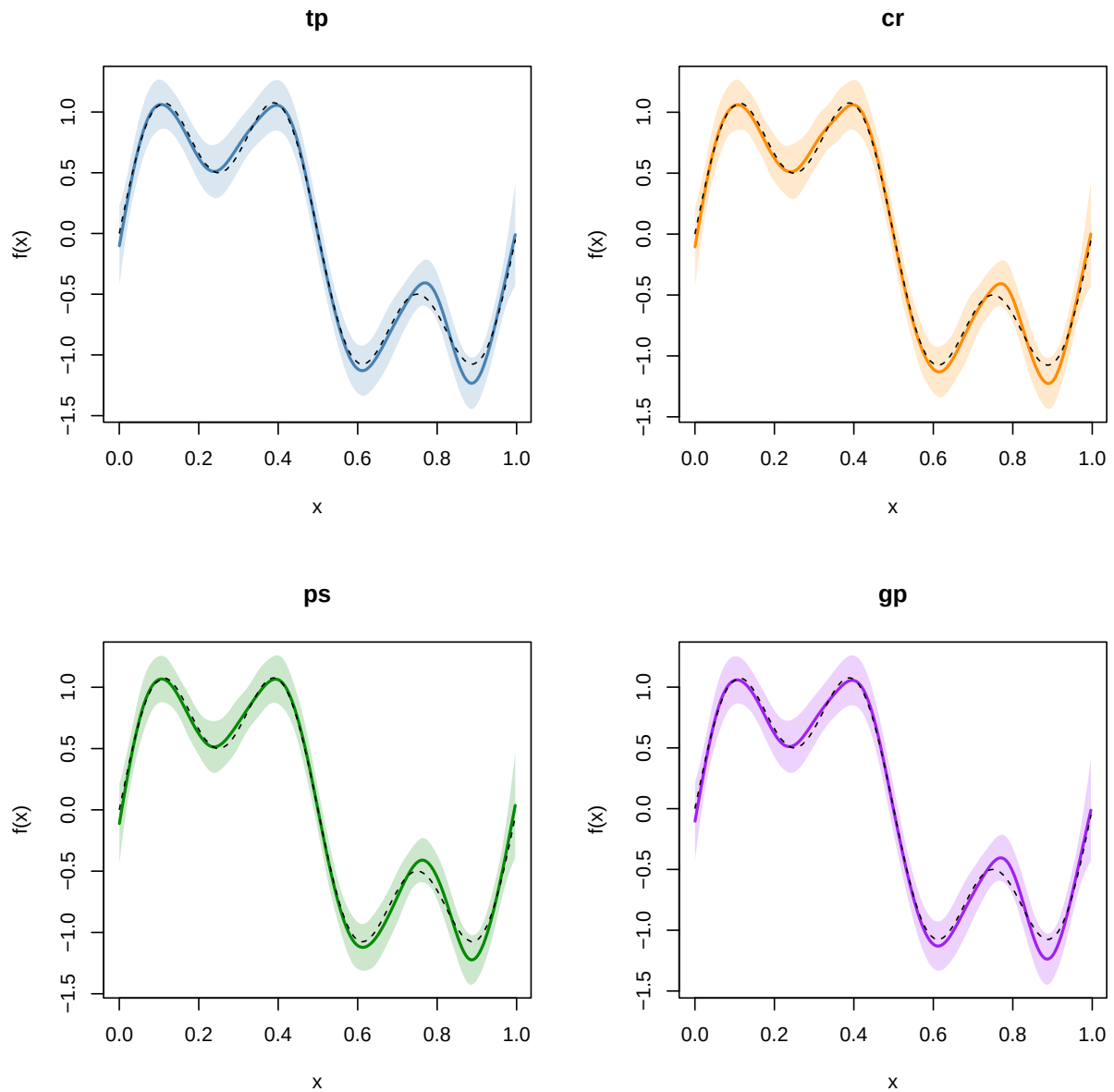
```
se_list <- lapply(bases, function(bs) {  
  se <- smooth_estimates(models[[bs]], n = 200)  
  se$basis <- bs  
  se  
})  
se_all <- do.call(rbind, se_list)  
  
ggplot(se_all, aes(x = x, y = .estimate, colour = basis)) +  
  geom_line(linewidth = 1) +  
  geom_line(data = data.frame(x = x, y = f_true),  
            aes(x = x, y = y), inherit.aes = FALSE,  
            linetype = "dashed", linewidth = 1) +  
  labs(x = "x", y = "f(x)", title = "Smooth estimates by basis type",  
        colour = "Basis") +  
  theme_minimal()
```



Each basis with its confidence band:

```
par(mfrow = c(2, 2))
colors <- c("steelblue", "darkorange", "green4", "purple")

for (i in seq_along(bases)) {
  bs <- bases[i]
  se <- smooth_estimates(models[[bs]], n = 200)
  plot(se$x, se[[".estimate"]], type = "l", lwd = 2, col = colors[i],
       ylim = range(c(se[[".estimate"]] - 2 * se[[".se"]], se[[".estimate"]] + 2 * se[[".se"]]),
       xlab = "x", ylab = "f(x)", main = bs)
  polygon(c(se$x, rev(se$x)),
         c(se[[".estimate"]] - 2 * se[[".se"]], rev(se[[".estimate"]] + 2 * se[[".se"]])),
         col = adjustcolor(colors[i], alpha.f = 0.2), border = NA)
  lines(x, f_true, lty = 2, lwd = 1)
}
```



```
par(mfrow = c(1, 1))
```

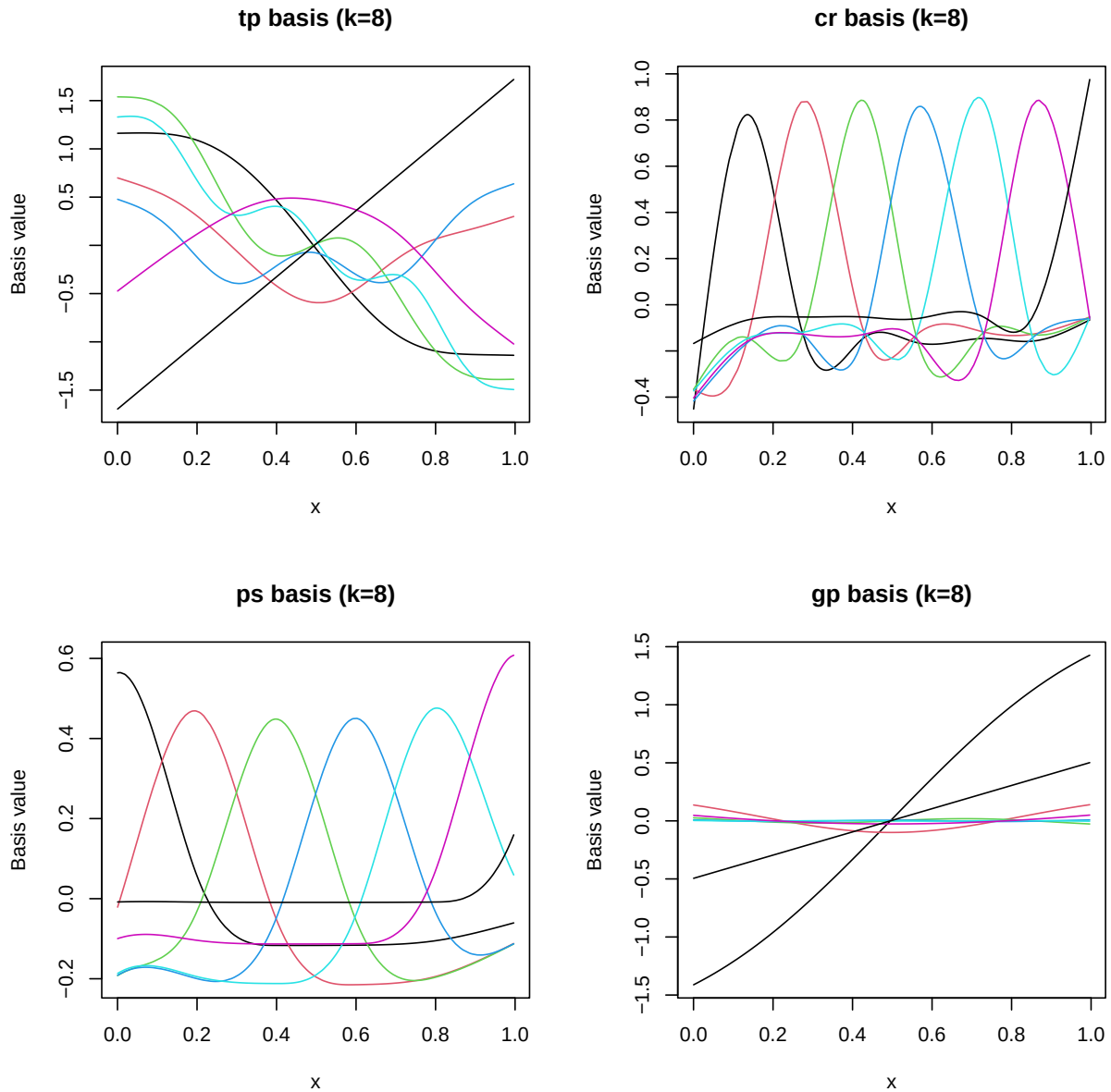
Visualizing basis functions

To understand how each basis works, we can examine the model matrix columns for a small number of basis functions ($k = 8$):

```

par(mfrow = c(2, 2))
for (i in seq_along(bases)) {
  bs <- bases[i]
  m_small <- gam(y ~ s(x, k = 8, bs = bs), data = df, method = "REML")
  X <- model.matrix(m_small)
  # Remove intercept column
  X_smooth <- predict(m_small, type = "lpmatrix")[, -1]
  k_cols <- ncol(X_smooth)
  ord <- order(df$x)
  matplot(df$x[ord], X_smooth[ord, ], type = "l", lty = 1,
          xlab = "x", ylab = "Basis value",
          main = paste0(bs, " basis (k=8)"))
}

```



```
par(mfrow = c(1, 1))
```

When to use which basis

- **Thin plate ("tp")**: The default choice. Works well in any dimension. Optimal smoothness in a certain mathematical sense. Slightly more expensive than knot-based alternatives for large datasets.

- **Cubic regression spline ("cr"):** Efficient for 1D smoothing with knots at data quantiles. Produces smooth curves that are natural cubic splines. Good default for univariate smooths.
- **P-spline ("ps"):** B-spline basis with a difference penalty on adjacent coefficients. Evenly spaced knots. Computationally efficient and well-behaved, especially for evenly sampled data.
- **Gaussian process ("gp"):** Uses a squared-exponential covariance kernel. Produces very smooth curves. Useful when the underlying function is believed to be infinitely differentiable.

Summary

In this vignette we:

1. Simulated bumpy data with both low- and high-frequency components
2. Fitted GAMs using four different basis types (TP, CR, PS, GP)
3. Compared EDF, deviance, and smooth estimates across bases
4. Visualized the raw basis functions for each type
5. Discussed when each basis is most appropriate

The next vignette demonstrates models with multiple smooth terms.